Improving Knowledge Distillation via Head and Tail Categories

Liuchi Xu, Jin Ren, Zhenhua Huang, Weishi Zheng, *Member, IEEE*, and Yunwen Chen, *Member, IEEE*

***Abstract*— Knowledge distillation (KD) is a technique that transfers "dark knowledge" from a deep teacher network (teacher) to a shallow student network (student). Despite significant advances in KD, existing work has not adequately mined two crucial types of knowledge: 1) the knowledge of head categories, which represents the relationship between the target category and its similar categories. Our findings reveal that this highly similar (complex) knowledge is essential for improving student's performance; and 2) the effectively utilized knowledge of tail categories. Existing studies often treat the non-target categories collectively without sufficiently considering the effectiveness of knowledge from tail categories. To tackle these challenges, we reformulate classical KD (ReKD) into two components: Top-K Inter-class Similar Distillation (TISD) and Non-Top-K Inter-class Discriminability (NTID). Firstly, TISD captures and imparts the knowledge of head categories to the student. Our experimental results have verified that TISD is particularly effective in transferring the knowledge of head categories, even in fine-grained dataset classification. Secondly, we theoretically show that the weighting coefficient of NTID increases with the probability of Top-K, leading to stronger suppression of knowledge transfer for tail categories. This observation explains why difficult samples are more informative than simple ones. To better utilize both types of knowledge, we optimize both TISD and NTID using different weighting coefficients, thereby enhancing the student's ability to learn this valuable knowledge from both head and tail categories. Furthermore, our extensive experimental results demonstrate that ReKD achieves state-of-the-art performance on various image classification datasets, including CIFAR-100, Tiny-ImageNet, and ImageNet-1K, as well as object detection and instance segmentation using the MS-COCO dataset.**

***Index Terms*— Knowledge distillation, knowledge of head categories, knowledge of tail categories, image classification, object detection, instance segmentation.**

I. Introduction

LARGE-SCALE deep neural networks have overwhelmingly achieved state-of-the-art performance in computer vision tasks, including image classification [1], [2], [3], [4], [5], object detection [6], [7], [8], [9], and semantic segmentation [10], [11], [12]. However, these high-performance networks generally come with a substantial number of parameters and computational overhead, hindering their deployment on resource-constrained embedded devices, such as mobile phones and edge cameras [13]. Knowledge distillation (KD) [14] is one of the popular methods for obtaining efficient networks. Obtained networks can be readily implemented on embedded devices while effectively reducing memory and computational requirements [15].

The core idea of KD [14] transfers the "dark knowledge" from the logits layer of a complex teacher network (teacher) to a compact student network (student). Generally, KD transfers dark knowledge in the logits layer using softmax with temperature. The goal is to minimize the Kullback-Leibler (KL) divergence between their probability distributions. The logits layer provides the advantage of highly abstract semantic information representations, requiring less memory and having lower computational costs [15]. However, logit distillation may not achieve the desirable generalization performance, posing challenges when applying KD in various scenarios.

Several previous studies [16], [17], [18], [19], [20], [21] have demonstrated that the effectiveness of logits-based knowledge distillation significantly improves the student's performance via decoupling the classical KD loss. For instance, Zhao et al. [19] decompose classical KD into two parts: 1) binary prediction for the target and non-target categories knowledge distillation; and 2) the non-target categories knowledge distillation (excluding the target category "tabby" in Fig.1). Their work provides theoretical insights into the limitations of KD, highlighting the high coupling that hampers its effectiveness in transferring knowledge from non-target categories. Yang et al. [20] and Li et al. [21] separately transfer the knowledge of the target category and non-target categories to the student. Ding et al. [18] discard target category knowledge and adopt non-target categories knowledge as an optimization objective. While these studies [18], [19], [20], [21] have made efforts to investigate logit distillation for improved

Manuscript received 28 May 2023; revised 16 August 2023; accepted 14 October 2023. Date of publication 19 October 2023; date of current version 9 May 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62172166 and Grant 61772366 and in part by the Guangdong Basic and Applied Basic Research under Grant 2022A1515011380. *(Liuchi Xu and Jin Ren contributed equally to this work.) (Corresponding author: Zhenhua Huang.)*

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This article has supplementary material provided by the authors and color versions of one or more figures available at https://doi.org/10.1109/TCSVT.2023.3325814.

Digital Object Identifier 10.1109/TCSVT.2023.3325814

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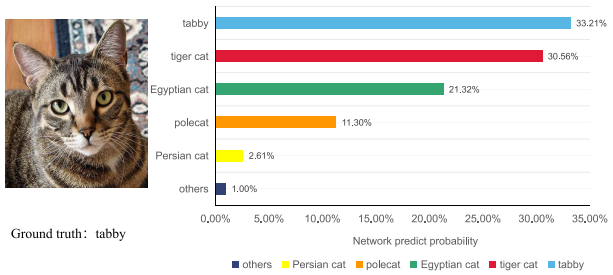


Fig. 1. The target category predicted by the network is "tabby" for Top-1 on the ImageNet-1K dataset.

performance, they failed to consider the relationship between the target and its similar non-target categories. Overlooking this relationship may potentially result in a deficiency of discriminative information, making it challenging to distinguish these categories. Consequently, the student may not fully learn the discriminative feature representations from the teacher, resulting in suboptimal performance.

Recognizing the similarity of knowledge between categories is undeniably significant. For instance, teachers demonstrating strong performance can easily identify confusing or highly similar knowledge compared to students. During the training phase, teachers emphasize this knowledge and encourage students to imitate it repeatedly. This approach deepens students' understanding, alleviates cognitive doubts and confusion, and enhances their ability to discriminate between similar categories. Therefore, acquiring this knowledge is essential for students to improve their accuracy effectively. As shown in Fig. 1, we define the Top-K categories with similarity as knowledge of head categories at the logit level. For example, when K=2, "tabby" and "tiger cat" are considered knowledge of head categories, while the remaining categories are classified as knowledge of tail categories. It is relatively simple to visually distinguish between "tabby" and "woman," as there is very little semantic similarity between them. However, the target category "tabby" and its similar categories display considerable visual similarity, because they share semantic features such as fur, eyes, noses, and postures in Fig. 1. Discerning similar categories poses a challenge for students with limited abilities. By capturing the head categories, noisy categories can be filtered out. Therefore, to facilitate student absorption and enhance classification performance, the teacher should focus on transferring these more relevant feature representations and discriminative information through knowledge of head categories. However, existing studies often disregard the importance of transferring knowledge of head categories.

Besides, we discover that with a fixed temperature, in fact, classical KD captures the knowledge of head categories with inter-class similarity information while neglecting tail category one (the remaining non-target categories knowledge) [22]. As depicted in Fig. 1, the knowledge of head categories typically occupies a significant portion of the overall knowledge, hindering knowledge transfer for tail categories [22]. Moreover, classical KD tends to discard a significant amount of knowledge of tail categories, potentially resulting in suboptimal performance. Therefore, it is crucial to consider the knowledge of tail categories for effective distillation [19]. However, classical KD often ignores the knowledge of tail categories when it comes to effective distill.

In this work, we propose a theoretical reformulation of classical KD, namely as **ReKD**, which consists of two components:**T**op-K **I**nter-class **S**imilar **D**istillation (**TISD**), and **N**on-**T**op-K **I**nter-class **D**iscriminability (**NTID**). Our objective is to present a simple yet effective knowledge distillation method that captures both the knowledge of head categories with inter-class similarity and knowledge of tail categories with the remaining categories at the logit level. TISD captures the knowledge of head categories from the teacher and imparts it to the student. Specifically, TISD focuses on grasping the knowledge of head categories based on the Top-K predicted probability. By emphasizing this aspect of knowledge during distillation, students acquire a deeper understanding, enhancing their ability to discriminate between these categories and alleviating potential doubts regarding their classification.

The NTID component delves deeply into the knowledge of tail categories to facilitate its transfer more effectively. Our theoretical findings suggest that as the probability of Top-K increases, the NTID weighting coefficient decreases. This decrease in the weight coefficient may result in a more severe suppression of the knowledge of tail categories, thereby hindering its transfer. Besides, the NTID weighting coefficient reveals that training samples with simpler classification tasks have a higher probability of being in the Top-K, which can lead to greater suppression of knowledge transfer for tail categories. This finding explains that simple training samples provide more limited knowledge compared to difficult ones.

To effectively improve the student's classification performance, it is crucial for the student to learn knowledge from both head and tail categories. By incorporating these two types of knowledge, we can ensure a more balanced optimization of the loss. To achieve this balance, we utilize different weighting coefficients to balance their respective losses. It enables the student to benefit from both knowledge of head and tail. Our proposed approach primarily focuses on logit distillation and can be combined with feature distillation to further enhance the performance of the student.

Overall, our contributions are summarized as follows:

1) We present a novel perspective on KD, highlighting the significance of the knowledge of head and tail categories. This perspective provides insights into the potential of utilizing both types of knowledge in distillation.
2) We design the TISD component through decoupling classical KD. TISD effectively transfers knowledge of head categories from teacher to student. It enhances the student's ability to distinguish between similar categories. It is worth noting that classical KD might not fully utilize the knowledge of tail categories to improve students' performance.
3) We conduct a theoretical analysis of the NTID weighting coefficient and uncover a limitation of classical KD: the suppression of knowledge of tail categories becomes more severe as the Top-K probability increases. To address this issue, we introduce a hyperparameter to replace the NTID weighting coefficient via decoupling

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KD. This approach enables the student to learn knowledge of tail categories from the teacher flexibly.

4) Our proposed ReKD based on logit distillation achieves state-of-the-art performance. It demonstrates superior results in image classification tasks on the CIFAR-100, Tiny-ImageNet, CINIC-10, SVHN, and ImageNet-1K, as well as in object detection and instance segmentation tasks on the MS-COCO dataset.

Section II introduces the related work. Section III reviews the background of KD and existing decoupled KD methods. Section IV presents our proposed ReKD through theoretical analysis, including TISD and NTID components. Section V validates the performance of ReKD on different datasets and tasks. Section VI concludes the paper.

II. Related Work

A. Logits-Based Distillation

1) Understanding of Knowledge Distillation: Hinton et al. [14] come up with knowledge distillation (KD), an efficient method for transferring "dark knowledge" from a teacher to a student using soft labels [16], [17], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33]. Recent researchers have investigated KD's role to gain a deeper understanding. It has been observed that KD can be interpreted as a label smoothing technique (LST) [27]. Tang et al. [28] provide theoretical evidence that LST can improve student performance. Muller et al. [31] demonstrate that LST reduces differences between samples within the same class, enabling the student to learn more compact feature representations. Shen et al. [32] reveal that tight clustering significantly enhances the separability of semantically similar representations across different classes. Zhou et al. [29] investigate the role of soft labels and observe that soft label weights are provided by different samples from the perspective of bias and variance. Cho et al. [34] conclude that a teacher's higher accuracy does not guarantee higher accuracy for a student. Stanton et al. [30] highlight a key optimization challenge that causes significant prediction mismatches between the teacher and the student. Furlanello et al. [25] propose a retraining process utilizing regenerative networks to explore knowledge transfer between models of equal complexity. They discover that the student's classification performance in KD can outperform the teacher's. Zhang et al. [23] also analyze the role of KD by quantifying the amount of knowledge.

2) Distilling Soft Labels: Although KD has been extensively understood, its effectiveness has not been fully studied. Our work primarily focuses on the final output layer of the network and investigates the form of logits-based distillation known as decoupled labels. Previous studies have paid attention to investigating the classical KD loss. For instance, Yang et al. [20] reformulate the classical KD loss, decomposing it and studying its relationship with the cross-entropy (CE) loss. They also reveal that classical KD can be considered a combination of the CE loss, maintaining the same form as the original CE loss, and a distributed loss. This CE loss decoupled from KD does not bring new knowledge from the teacher. Li et al. [21] decompose classical KD into correction guidance, label smoothing, and class discriminability. These three components can improve classical KD performance. The above-mentioned methods have achieved state-of-the-art results in image classification. However, these methods often consider the non-target categories as a whole when decoupling classical KD, overlooking the relationship between the target and its similar non-target categories.

In our work, we further decompose classical KD into two components: TISD and NTID. These components consider both the knowledge of head and tail categories, reducing information loss. The teacher emphasizes the knowledge of head categories and imparts relevant feature representations and discriminative knowledge to the students. Meanwhile, we adopt α and β as substitutes for their decoupling weighting coefficients. The student flexibly learns both the knowledge of head and tail categories simultaneously.

B. Features-Based Distillation

Several studies [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50] primarily focus on transferring knowledge from intermediate features, which contain richer information and more diverse knowledge. This diverse knowledge allows students to learn valuable knowledge more effectively. For example, FitNet [51] utilizes Mean Squared Error (MSE) to allow the student to mimic the teacher's global intermediate features. AT [39] adopts an attention mechanism to guide knowledge transfer from intermediate features. This helps the student's intermediate features to approximate those of the teacher. ICKD [41] explores channel diversity knowledge and preserves channel-related information for knowledge distillation. CRD [37] proposes contrastive learning for knowledge distillation. It introduces contrastive targets to facilitate knowledge transfer from the teacher's representations to the student. VID [52] maximizes the lower bound of mutual information between the teacher and the student through variational distillation. ReviewKD [53] regards shallow knowledge as old knowledge and performs continuous reviews. Besides, RKD [35] is an approach focusing on transferring relationship knowledge between samples rather than the knowledge of a single sample. To a large extent, these methods continue to enhance the teacher's feature extraction ability. However, they do not specify which layer of features is most effective in facilitating knowledge transfer from the teacher to the student. Features-based distillation methods do not improve classical KD and incur significant computational and storage overhead. Our approach can be combined with features-based methods to improve the student's performance.

III. Background

In this section, we provide a brief overview of the classical knowledge distillation (KD) method. We then review existing methods for decoupling classical KD.

A. Classical Knowledge Distillation

Given a mini-batch of training samples $\chi = \{(x_i, y_i)\}_{i=1}^{n} \subseteq \mathcal{D} = (\mathcal{D}^{train}, \mathcal{D}^{val})$, where n and $\mathcal{D}$ denote the batch size of

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training sample and the entire dataset size, respectively. The backbone network $\mathcal{F}$ is parameterized by θ, and it produces the output $\mathcal{F}(x_i;\theta)$ for a given training sample x_i. The temperature coefficients for the teacher and student are denoted by $\mathcal{T}_t$ and $\mathcal{T}_s$, respectively. For a training sample x_i with label y_i, we define the probability distribution and logit of the k-th class for the teacher as $\mathcal{P}_i^t(k) \in [\mathcal{P}_i^t(1), \mathcal{P}_i^t(2), \cdots, \mathcal{P}_i^t(C)] \in \mathbb{R}^{1\times C}$ and $\mathcal{F}_i^t(x_i;k;\theta)$, respectively. Similarly, we define the probability distribution and logit of the k-th class for the student as $\mathcal{P}_i^s(k) \in [\mathcal{P}_i^s(1), \mathcal{P}_i^s(2), \cdots, \mathcal{P}_i^s(C)] \in \mathbb{R}^{1\times C}$ and $\mathcal{F}_i^s(x_i;k;\theta)$, respectively. The probability distributions of the teacher and student for the k-th category are calculated as follows:

$$\mathcal{P}_i^t(k) = \frac{exp(\mathcal{F}_i^t(x_i;k;\theta)/\mathcal{T}_t)}{\sum_{j=1}^{C} exp(\mathcal{F}_j^t(x_i;k;\theta)/\mathcal{T}_t)}, \tag{1}$$

$$\mathcal{P}_i^s(k) = \frac{exp(\mathcal{F}_i^s(x_i;k;\theta)/\mathcal{T}_s)}{\sum_{j=1}^{C} exp(\mathcal{F}_j^s(x_i;k;\theta)/\mathcal{T}_s)}, \tag{2}$$

where the variable C denotes the number of categories. The temperature coefficients for the teacher and student are represented by $\mathcal{T}_t$ and $\mathcal{T}_s$, both equal to $\mathcal{T}$, respectively.

In classical KD, the technique commonly utilizes Kullback-Leibler (KL) divergence as the loss metric for distillation. The objective of training the student is to approximate the probability distribution of the teacher and minimize the discrepancy between their distributions. The classical KD loss can be expressed as follows:

$$\mathcal{L}_{KD} = \frac{1}{N}\sum_{j=1}^{N}\sum_{k=1}^{C} \mathcal{P}_j^t(k)\log(\frac{\mathcal{P}_j^t(k)}{\mathcal{P}_j^s(k)}), \tag{3}$$

where $\mathcal{P}_j^t(k)$ and $\mathcal{P}_j^s(k)$ denote the probability of the k-th category in the j-th training sample for the teacher and student, respectively. N denotes the number of training samples. We omit the temperature coefficient $\mathcal{T}$ for simplicity.

The original cross-entropy loss for image classification in the student can be formulated as follows:

$$\mathcal{Q}_j^s(k) = \frac{exp(\mathcal{F}_j^s(x_j;k;\theta))}{\sum_{i=1}^{C} exp(\mathcal{F}_i^s(x_j;k;\theta))}, \tag{4}$$

$$\begin{aligned}\mathcal{L}_{CE} &= -\frac{1}{N}\sum_{j=1}^{N}\sum_{k=1}^{C} y_j^s(k)\log \mathcal{Q}_j^s(k)) \\ &= -\frac{1}{N}\sum_{j=1}^{N}\sum_{k=1}^{C} \log(\mathcal{Q}_j^s(k)),\end{aligned} \tag{5}$$

where $y_j^s(k)$ represents the one-hot label for the k-th category in the j-th training sample, and $\mathcal{Q}_j^s(k)$ represents the probability of the k-th class in the j-th training sample for student with temperature 1.0.

The total loss $\mathcal{L}_{total}$ for knowledge distillation consists of two parts: the classical KD loss $\mathcal{L}_{KD}$ and the original cross-entropy loss $\mathcal{L}_{CE}$. Mathematically, the total loss can be expressed as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{CE} + \lambda\mathcal{L}_{KD}, \tag{6}$$

where λ is a tuned hyperparameter for balancing both losses.

B. Decoupling Classical KD Loss

For simplicity, we have removed N from Eq. (3), where N represents the number of training samples. It is worth noting that $\mathcal{P}^t(k)$ and $\mathcal{P}^s(k)$ represent the probability of the k-th category for a training sample, as predicted by the teacher and student, respectively.

1) Decoupling Classical KD Loss in [20]: According to Eq. (3), the simplified representation of the $\mathcal{L}_{KD}$ loss is given as follows:

$$\begin{aligned}\mathcal{L}_{KD} &= -\sum_{k=1}^{C}\mathcal{P}^t(k)\log(\mathcal{P}^s(y) * \frac{\mathcal{P}^s(k)}{\mathcal{P}^s(y)}) \\ &= -\sum_{k=1}^{C}\mathcal{P}^t(k)\log(\mathcal{P}^s(y)) - \sum_{k=1}^{C}\mathcal{P}^t(k)\log\frac{\mathcal{P}^s(k)}{\mathcal{P}^s(y)}).\end{aligned} \tag{7}$$

Given that $\sum_{k=1}^{C}\mathcal{P}^t(k) = \sum_{k=1}^{C}\mathcal{P}^s(k) = 1.0$ according to Eqs. (1) and (2), the second term in Eq. (7) can be further simplified. When $k = y$ (where y represents the predicted target category), we have $\mathcal{P}^t(y)\log\frac{\mathcal{P}^s(y)}{\mathcal{P}^s(y)} = 0$. Therefore, Eq.(7) can be reformulated as follows:

$$\mathcal{L}_{KD} = -\log(\mathcal{P}^s(y)) - \underbrace{\sum_{k\neq y}^{C}\mathcal{P}^t(k)\log\frac{\mathcal{P}^s(k)}{\mathcal{P}^s(y)})}_{\text{Non-Target Knowledge Distillation}}. \tag{8}$$

We can observe that -log($\mathcal{P}^s(y)$) shares the same form as Eq. (5) based on Eq. (8), indicating that it does not provide additional knowledge to the student. Besides, the second term of Eq. (8) focuses on distilling the non-target categories as a whole, disregarding their relationship with the target category.

2) Decoupling Classical KD Loss in [21]: Some researchers have analyzed the classical KD loss and decomposed it into correct guidance, label smoothing, and class discrimination. They found that all three factors contribute to effective knowledge transfer. The $\mathcal{L}_{KD}$ formula, as derived from Eq. (3), can be expressed as follows:

$$\begin{aligned}\mathcal{L}_{KD} = &\underbrace{-\mathcal{P}^t(y)\log\mathcal{P}^s(y)}_{\text{Correct Guidance}} - \underbrace{\sum_{k\neq y}^{C} e(\mathcal{P}^t)\log\mathcal{P}^s(k)}_{\text{Smooth Regularization}} \\ &- \underbrace{\sum_{k\neq y}^{C}(\mathcal{P}^t(k) - e(\mathcal{P}^t))\log\mathcal{P}^s(k)}_{\text{Class Discriminability}},\end{aligned} \tag{9}$$

where we define $e(\mathcal{P}^t)) = \frac{1}{C-1}\sum_{k=1,k\neq y}^{C}\mathcal{P}^t(k)$.

The second and third terms of Eq. (9) indicate that the weights of non-targets differ from those of the source during distillation. However, the relationship between non-targets and the target category remains independent.

3) Decoupling Classical KD Loss in [19]: Some researchers have proposed decomposing KD into two separate components: target knowledge distillation and non-target knowledge distillation. The detailed formula

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expression for this approach is as follows:

$$\mathcal{L}_{KD} = (1 - P^t(y)) \underbrace{\sum_{k \neq y}^{C} \mathcal{P}^t(k) \log \frac{\mathcal{P}^s(k)}{\mathcal{P}^s(y)})}_{\text{Non-Target Knowledge Distillation}} + \underbrace{P^t(y) \log \frac{P^t(y)}{P^s(y)} + \bar{P}^t(\bar{y}) \log \frac{\bar{P}^t(\bar{y})}{\bar{P}^s(\bar{y})}}_{\text{Target Class Knowledge Distillation}}, \quad (10)$$

where $P^t(y)$ and $\bar{P}^t(\bar{y})$ represent the sum of the target and all non-target probabilities, respectively.

From the analysis of Eq. (10), it can be observed that the probability $P^t(y)$ is higher, and the weight of non-target knowledge distillation decreases. This suggests a more significant suppression of non-target distillation. Moreover, some studies have observed that these non-target categories tend to neglect their relationship with the target category.

In general, based on Eqs. (8-10), it can be observed that existing studies [19], [20], [21] have overlooked the relationship between the target category and non-target categories.

IV. Proposed Method

In this section, we first provide a theoretical discussion regarding the decoupled logits from the head and tail categories. We then describe our method ReKD, including TISD and NTID components, as illustrated in Fig. 2. Finally, we optimize the algorithm and provide code implementation details.

A. Decoupling Logits From Head and Tail Categories

Let the vector $\mathbf{z} = [z_1, z_2, \ldots, z_C] \in \mathbb{R}^{1\times C}$, where z_i is the logit of the network for the i-th class. The output of network is parameterized by $\mathcal{F}(x_i; C; \theta)$, where C is the total number of categories. The classification probability, obtained by applying the softmax function to the network output, can be expressed as follows:

$$p_i = \frac{exp(z_i)}{\sum_{j=1}^{C} exp(z_j)}, \quad (11)$$

where z_i represents the logit of the i-th class.

We define P_{TopK} as the sum of probabilities of the top k predicted categories. $K=\{Index\}_{i=1}^{k}$ represents the index set of the top k predictions with the highest accuracy. We also define $\bar{P}_{TopK}$ as the sum of probabilities of the remaining categories. They can be mathematically expressed as follows:

$$P_{TopK} = \frac{\sum_{i=1,i\in K}^{C} exp(z_i)}{\sum_{j=1}^{C} exp(z_j)}, \quad \bar{P}_{TopK} = \frac{\sum_{i=1,i\notin K}^{C} exp(z_i)}{\sum_{j=1}^{C} exp(z_j)}. \quad (12)$$

We define $Q_{TopK}(i)$ as the probability of the i-th category within the K categories, and $\bar{Q}_{TopK}(i)$ as the probability of the i-th category within the remaining categories. They can be mathematically expressed as follows:

$$Q_{TopK}(i) = \frac{exp(z_i)}{\sum_{j=1,j\in K}^{C} exp(z_j)}, \quad \bar{Q}_{TopK}(i) = \frac{exp(z_i)}{\sum_{j=1,j\notin K}^{C} exp(z_j)}. \quad (13)$$

Based on the preceding equations, we can derive the probabilities p_i for the network predictions that belong to the K categories as $p_i = P_{TopK} Q_{TopK}(i)$ according to Eq. (12) and Eq. (13). For network predictions that do not belong to the K categories, we have $p_i = \bar{P}_{TopK} \bar{Q}_{TopK}(i)$ according to Eq. (12) and Eq. (13). That is,

$$p_i = \begin{cases} P_{TopK} Q_{TopK}(i), & i \in K, \\ \bar{P}_{TopK} \bar{Q}_{TopK}(i), & i \notin K. \end{cases} \quad (14)$$

Now, we reformulate the distillation loss based on the aforementioned knowledge. The comprehensive formula representation is provided as follows:

$$\mathcal{L}_{KD} = \sum_{i=1}^{C} p_i^t \log(\frac{p_i^t}{p_i^s}) = \sum_{i=1,i\in K}^{C} p_i^t \log(\frac{p_i^t}{p_i^s}) + \sum_{i=1,i\notin K}^{C} p_i^t \log(\frac{p_i^t}{p_i^s}). \quad (15)$$

On the one hand, we can analyze the first term of Eq. (15) by substituting $p_i = P_{TopK} Q_{TopK}(i)$ into it, according to Eq. (14). For convenience, let us denote the first term of Eq. (15) as Δ. The detailed formula can be expressed as follows:

$$\begin{aligned} \Delta &= \sum_{i=1,i\in K}^{C} P_{TopK}^t Q_{TopK}^t(i) \log \frac{P_{TopK}^t Q_{TopK}^t(i)}{P_{TopK}^s Q_{TopK}^s(i)} \\ &= P_{TopK}^t \log \frac{P_{TopK}^t}{P_{TopK}^s} \sum_{i=1,i\in K}^{C} Q_{TopK}^t(i) \\ &\quad + P_{TopK}^t \sum_{i=1,i\in K}^{C} Q_{TopK}^t(i) \log \frac{Q_{TopK}^t(i)}{Q_{TopK}^s(i)}. \end{aligned}$$

Since $\sum_{i=1,i\in K}^{C} Q_{TopK}^t(i) = 1$, the Δ simplifies to its final form as follows:

$$\Delta = P_{TopK}^t \log \frac{P_{TopK}^t}{P_{TopK}^s} + P_{TopK}^t KL(Q_{TopK}^t || Q_{TopK}^s), \quad (16)$$

where

$$KL(Q_{TopK}^t || Q_{TopK}^s) = \sum_{i=1,j\in K}^{C} Q_{TopK}^t(i) \log \frac{Q_{TopK}^t(i)}{Q_{TopK}^s(i)}.$$

On the other hand, we can analyze the second term of Eq. (15) by substituting $p_i = \bar{P}_{TopK} \bar{Q}_{TopK}(i)$ into it, according to Eq.(14). For convenience, let us denote the second term of Eq. (15) as Λ. The detailed formula can be expressed as follows:

$$\begin{aligned} \Lambda &= \sum_{i=1,i\notin K}^{C} \bar{P}_{TopK}^t \bar{Q}_{TopK}^t(i) \log \frac{\bar{P}_{TopK}^t \bar{Q}_{TopK}^t(i)}{\bar{P}_{TopK}^s \bar{Q}_{TopK}^s(i)} \\ &= \bar{P}_{TopK}^t \log \frac{\bar{P}_{TopK}^t}{\bar{P}_{TopK}^s} \sum_{i=1,i\notin K}^{C} \bar{Q}_{TopK}^t(i) \end{aligned}$$

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$$+\sum_{i=1,i\notin K}^{C}\bar{P}^t_{TopK}\bar{Q}^t_{TopK}(i)\log\frac{\bar{Q}^t_{TopK}(i)}{\bar{Q}^s_{TopK}(i)}.$$

Similarly, we observe that $\sum_{i=1,i\notin K}^{C}\bar{Q}^t_{TopK}(i)=1$. Thus, the Λ can be simplified to its final form as follows:

$$\Lambda=\bar{P}^t_{TopK}\log\frac{\bar{P}^t_{TopK}}{\bar{P}^s_{TopK}}+(1-P^t_{TopK})KL(\bar{Q}^t_{TopK}||\bar{Q}^s_{TopK}), \tag{17}$$

where

$$KL(\bar{Q}^t_{TopK}||\bar{Q}^s_{TopK})=\sum_{i=1,i\notin K}^{C}\bar{Q}^t_{TopK}(i)\log\frac{\bar{Q}^t_{TopK}(i)}{\bar{Q}^s_{TopK}(i)}.$$

Combining Eq. (15), Eq. (16), and Eq.(17), the final decoupled KD distillation loss can be expressed as follows:

$$\begin{aligned}\mathcal{L}_{KD}=&P^t_{TopK}\underbrace{KL(Q^t_{TopK}||Q^s_{TopK})}_{TISD}\\&+(1-P^t_{TopK})\underbrace{KL(\bar{Q}^t_{TopK}||\bar{Q}^s_{TopK})}_{NTID}\\&+\underbrace{P^t_{TopK}\log\frac{P^t_{TopK}}{P^s_{TopK}}+(1-P^t_{TopK})\log\frac{1-P^t_{TopK}}{1-P^s_{TopK}}}_{BCL}.\end{aligned} \tag{18}$$

We have observed several differences between Eq. (18) and Eqs. (8-10) as outlined below: 1) When K=1, Eq. (18) degenerates into DKD, as illustrated in Eq. (10); 2) The weight assigned to NTID differs from DKD when $K\geq 2$; 3) We further decouple another term, referred to as Binary Classification Loss (BCL) for P^t_{TopK} and $\bar{P}^t_{TopK}$.

From an intuitive perspective, as the value of K increases, the sum of probabilities for the top categories approaches 1.0, while the sum of probabilities for the tail categories approaches 0. This implies that the BCL loss becomes minimal, allowing us to ignore it for efficient computation and stable optimization. To establish the relationship in magnitude between the first and second terms of Eq. (18) and the first and second terms of BCL, respectively. Therefore, we propose the following proposition.

Proposition 1: The following inequality holds:

$$\begin{cases}-\sum_{i=1}^{K}p^t_i\log p^s_i\geq-\log P^s_{TopK},\\-\sum_{i=K+1}^{C}\bar{p}^t_i\log\bar{p}^s_i\geq-\log\bar{P}^s_{TopK}.\end{cases} \tag{19}$$

Proof: We prove Proposition 1 by mathematical induction. The detailed proof can be found in the Supplementary Material.

We deduce from Eq. (19) that Proposition 1 suggests that the sum of the first two terms in Eq. (18) outweighs the third term. By combining Eq. (18) and Eq. (19), we obtain the upper bound $\mathcal{L}_{up}$ of the KD loss:

$$\begin{aligned}\mathcal{L}_{KD}\leq&2P^t_{TopK}\underbrace{KL(Q^t_{TopK}||Q^s_{TopK})}_{TISD}\\&+2(1-P^t_{TopK})\underbrace{KL(\bar{Q}^t_{TopK}||\bar{Q}^s_{TopK})}_{NTID}\\=&\mathcal{L}_{up}\end{aligned} \tag{20}$$

Therefore, $\mathcal{L}_{CE}+\lambda\mathcal{L}_{KD}\leq\mathcal{L}_{CE}+2\lambda\mathcal{L}_{up}=\mathcal{L}_{CE}+\rho\mathcal{L}_{up}$. For the convenience of description, we replace 2λ with ρ here, and ρ is a tuned hyperparameter for balancing $\mathcal{L}_{CE}$ and $\mathcal{L}_{up}$. To ensure efficient computation and stable training of the network, we use an upper bound $\mathcal{L}_{up}$ as the optimization loss for knowledge distillation.

Drawing upon insights gained from our analysis, we propose a novel knowledge distillation approach that reformulates the classical KD into TISD and NTID, which separately transfer the knowledge they excel at capturing. Specifically, when $K\geq 2$, P^t_{TopK} increases until it reaches 1.0, as observed in Eq. (20). Increasing P^t_{TopK} is beneficial for TISD but detrimental for NTID. Therefore, we introduce two adjustable parameters α and β to replace the weighting coefficients associated with P^t_{TopK} and $1-P^t_{TopK}$ in Eq. (20), respectively. The specific implementation of TISD and NTID can be found in Subsection IV-B1 and Subsection IV-B2, respectively. Our approach enables students to learn the knowledge of head and tail categories, effectively improving their performance during distillation. Their details are expressed as follows:

$$\mathcal{L}_{up}=\alpha\mathcal{L}_{TISD}+\beta\mathcal{L}_{NTID}. \tag{21}$$

B. Methodology Description

Existing studies have overlooked the relationship between the target category and non-target categories, which leads to the loss of discriminative information. Besides, the knowledge of head categories remains underutilized, resulting in suppressed and imbalanced knowledge transfer during training. To tackle these challenges, we propose the ReKD method which consists of two components: TISD and NTID. By utilizing these two components, we can effectively capture the knowledge of head categories with inter-class similarity and better transfer knowledge of tail categories. ReKD is based on logit distillation, which is simple to implement and does not require additional computational overhead, thus enabling efficient utilization of those two types of knowledge. Building upon the theoretical analysis discussed in Section IV-A, we elaborate on these methods as follows.

1) Knowledge of Head Categories Transfer in TISD: TISD can transfer the knowledge of head categories from the teacher and impart it to the student, allowing the students to absorb this knowledge and improve their performance effectively. We use the logits layer of the network as the target for distillation. To obtain the head categories (which are offered in the orange section of Fig. 2), we decouple the logits layer of the network. The implementation details of TISD are as follows: Let the student vector is $\mathbf{v_s}=[z^s_1,z^s_2,\cdots,z^s_K]\in\mathbb{R}^{1\times K}$, where $\mathbf{v_s}$ is the logits layer of the student, which is parameterized by

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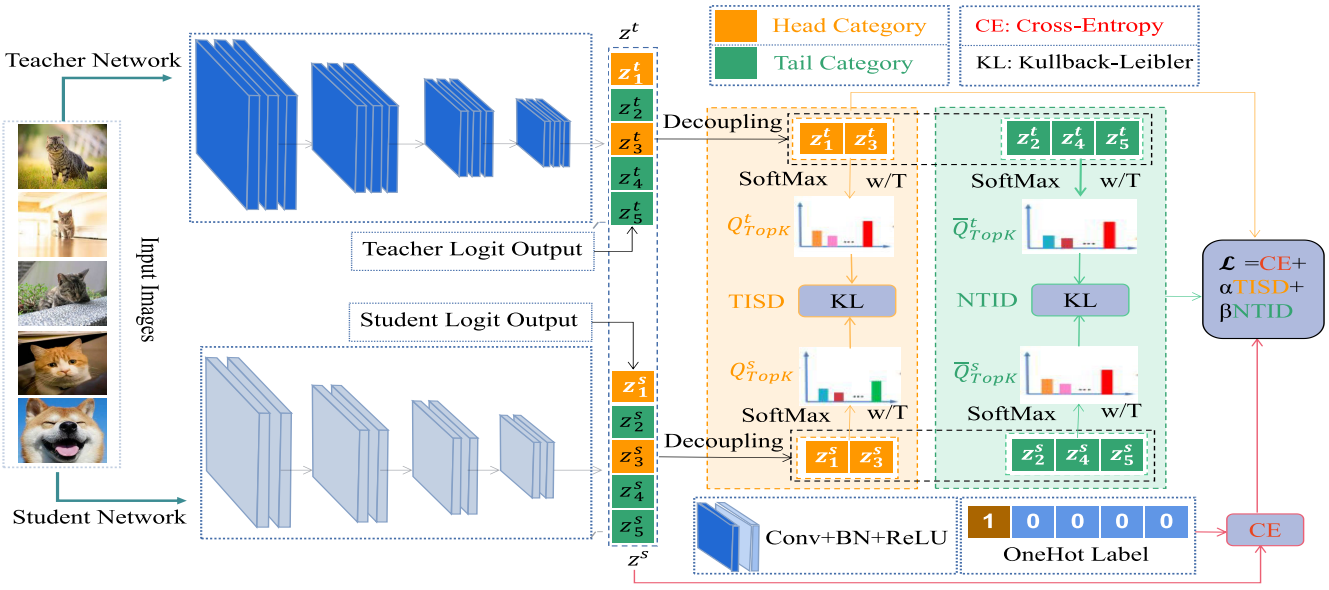


Fig. 2. The overview of our proposed framework. The top and bottom branches represent the teacher and student networks, respectively. We adopt the logits layer as a distillation objective. We propose the TISD component that can transfer knowledge of head categories from teacher to student by minimizing the Kullback-Leibler (KL) divergence between their probability distributions(indicated by the orange box). Additionally, we also propose the NTID component can transfer knowledge of tail categories from teacher to student (indicated by the green box).

$\mathcal{F}(x_i; K; \theta)$ and K represents Top-K category labels (obtained from the student). Similarly, let the teacher vector is $\mathbf{v_t} = [z_1^t, z_2^t, \cdots, z_K^t] \in \mathbb{R}^{1\times K}$. The distillation loss for TISD can be expressed as follows:

$$\begin{aligned} Q_{TopK}^t &= SoftMax(\mathbf{v_t}/\mathcal{T}), \\ Q_{TopK}^s &= SoftMax(\mathbf{v_s}/\mathcal{T}), \\ \mathcal{L}_{TISD} &= KL(Q_{TopK}^t || Q_{TopK}^s) \end{aligned} \tag{22}$$

$$= \sum_{i=1, i\in K}^{C} Q_{TopK}^t(i) \log\left(\frac{Q_{TopK}^t(i)}{Q_{TopK}^s(i)}\right), \tag{23}$$

where Q_{TopK}^t and Q_{TopK}^s represent the probability of the Top-K categories for the teacher and the student, respectively. $\mathcal{T}$ represents the temperature coefficient.

The comprehensive theoretical analysis for TISD is provided in Section IV-A, as detailed in Eq. (16) and Eq. (18).

2) Knowledge of Tail Categories Transfer in NTID: The tail categories possess a wealth of valuable knowledge that has not been fully utilized in existing research. To investigate the value of knowledge of tail categories, we propose the NTID component, which can effectively transfer knowledge of tail categories from the teacher to the student. We transfer this part of knowledge separately to avoid losing this information. As depicted in the green section of Fig. 2, we obtain the knowledge of tail categories by decoupling the logits layer of the network. Specifically, let the student vector be denoted as $\bar{\mathbf{v}}_\mathbf{s} = [z_{K+1}^s, z_{K+2}^s, \cdots, z_C^s] \in \mathbb{R}^{1\times(C-K)}$, where $\bar{\mathbf{v}}_\mathbf{s}$ is the logits layer of the student, which is parameterized by $\mathcal{F}(x_i; K; \theta)$, and K represents Top-K category labels (obtained from the student). In the same way, let the teacher vector be denoted as $\bar{\mathbf{v}}_\mathbf{t} = [z_{K+1}^t, z_{K+2}^t, \cdots, z_C^t] \in \mathbb{R}^{1\times(C-K)}$. The distillation loss for NTID can be expressed as follows:

$$\bar{Q}_{TopK}^t = SoftMax(\bar{\mathbf{v}}_\mathbf{t}/\mathcal{T}),$$

$$\bar{Q}_{TopK}^s = SoftMax(\bar{\mathbf{v}}_\mathbf{s}/\mathcal{T}), \tag{24}$$

$$\begin{aligned} \mathcal{L}_{NTID} &= KL(\bar{Q}_{TopK}^t || \bar{Q}_{TopK}^s) \\ &= \sum_{i=1, i\notin K}^{C} \bar{Q}_{TopK}^t(i) \log\left(\frac{\bar{Q}_{TopK}^t(i)}{\bar{Q}_{TopK}^s(i)}\right), \end{aligned} \tag{25}$$

where $\bar{Q}_{TopK}^t$ and $\bar{Q}_{TopK}^s$ represent the probability of the Non-Top-K categories for the teacher and the student, respectively. $\mathcal{T}$ represents the temperature coefficient.

The comprehensive theoretical analysis for NTID is provided in Section IV-A, as detailed in Eq. (17) and Eq. (18).

C. Optimization

We propose ReKD, which includes TISD and NTID components. The TISD component is designed to effectively transfer the knowledge of head categories from the teacher, with a particular emphasis on similarity to the target category. This ensures the transfer of discriminative information that is crucial for the student's learning. Simultaneously, the NTID component effectively transfers knowledge of tail categories to the student. During the training process, the teacher's parameters are frozen, and we train the student using both task-specific loss and distillation loss. Specifically, the former utilizes a cross-entropy loss $\mathcal{L}_{CE}$, and the latter uses distillation loss $\mathcal{L}_{DISTILL}$. The latter includes Top-K Inter-class Similar Distillation loss $\mathcal{L}_{TISD}$ and Non-Top-K Inter-class Discriminability loss $\mathcal{L}_{NTID}$. The total loss function, denoted as $\mathcal{L}_{TotalKD}$, can be represented as follows:

$$\mathcal{L}_{TotalKD} = \mathcal{L}_{CE} + \alpha\mathcal{L}_{TISD} + \beta\mathcal{L}_{NTID}, \tag{26}$$

where α and β are weight factors for balancing these losses. ReKD does not introduce additional hyperparameters and computational overhead. Algorithm 1 provides the pseudo-code of ReKD in a PyTorch-like [54] style for distillation, allowing for easy implementation and testing of ReKD.

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TABLE I

TEST ACCURACY (%) ON THE SVHN AND CINIC-10 DATASETS. **RESNET110** IS THE TEACHER, AND **RESNET20** IS THE STUDENT. TEACHER* REPRESENTS THE RESULT OF OUR IMPLEMENTATION. THE BEST RESULTS IN EACH EXPERIMENT ARE EMPHASIZED IN BOLD

Distillation Manner				Features				Logits			
Dataset	Student	Teacher	Teacher*	AT [39]	FitNet [51]	DML [16]	RKD [35]	KD [14]	SLKD [55]	ReKD	Δ
SVHN	94.98	96.63	96.72	95.81	95.97	96.11	96.03	95.43	96.25	**96.89**	+0.64
CINIC-10	82.43	86.45	84.25	82.84	82.93	83.05	82.98	82.58	83.16	**84.00**	+0.84

Algorithm 1 Pseudo Code of ReKD

```
#y_s: the logit of the student
#y_t: the logit of the teacher
#k: parameter for topk
#T: represents temperature coefficient
#alpha: the weight coefficient of TISD
#beta: the weight coefficient of NTID
alpha = opt.alpha
beta = opt.beta

#TISD distillation loss
s_mask = top_k_mask(y_s,k) #obtain head categories mask
#keeping the head categories labels
s_topk_logit = y_s / T-1000.0 * s_mask
t_topk_logit = y_t / T-1000.0 * s_mask
#normalization by row
s_topk_logit = F.log_softmax(s_topk_logit,dim=1)
t_topk_logit = F.softmax(t_topk_logit,dim=1)
L_TISD=F.kl_div(s_topk_logit,t_topk_logit,size_average=False)*
       (T * T) /y_s.size()[0]

#NTID distillation loss
not_s_mask=not_topk_mask(y_s,k)#obtain tail category mask
#keeping the tail category labels
s_not_topk_logit = y_s / T-1000.0 * not_s_mask
t_not_topk_logit = y_t / T-1000.0 * not_s_mask
#normalization by row
s_logsoftmax = F.log_softmax(s_not_topk_logit, dim=1)
t_softmax = F.softmax(t_not_topk_logit, dim=1)
L_NTID = F.kl_div(s_logsoftmax,t_softmax,size_average=False)*
       (T * T) /y_s.size()[0]
TotalKD =L_CE + alpha*L_TISD + beta*L_NTID #total loss
```

V. EXPERIMENTS

In this section, we first compare the performance of ReKD with state-of-the-art methods on various datasets to establish its superiority. We then conduct ablation studies to validate the effectiveness of the TISD and NTID components. Finally, we perform quantitative analysis to assess the effectiveness of ReKD. The datasets and experimental details can be found in Supplementary Material. We conduct three independent runs and report the Top-1 validation accuracy (%) of the student, calculated as the average results from the validation set (except for the MS-COCO2017 dataset).

A. Comparison With State-of-the-Art Methods

1) Image Classification Results on SVHN and CINIC-10: We conduct experiments and compare different methods on the SVHN and CINIC-10 datasets. Our experiments demonstrate that ReKD outperforms the state-of-the-art SLKD based on logit distillation, as shown in Table I. In our experiments, we employ ResNet110 as the teacher and ResNet20 as the student. ReKD achieves performance improvements of 0.64% and 0.84% on the SVHN and CINIC-10 datasets, respectively. These results indicate that ReKD is effective for these datasets with fewer categories.

2) Image Classification Results on CIFAR-100: We conduct extensive experiments on CIFAR-100 and present our results in Table II. We divide the methods into two groups: features-based and logits-based. Our proposed ReKD has achieved superior results at the logit distillation compared to other methods using the same model architecture. Specifically, in Table II, ReKD outperforms classical KD, with average gains of 1.53%, 1.26%, 1.21%, 3.65%, 1.21%, 1.30%, and 1.64%, in that order from left to right. These results demonstrate that ReKD is more effective for the same architecture on the CIFAR-100.

3) Image Classification Results on ImageNet-1K: We also conduct experiments on the more challenging ImageNet-1k dataset and present the results in Table III. For this evaluation, we employ ResNet34 as the teacher and ResNet18 as the student. ReKD achieves Top-1 and Top-5 performance improvements of 1.19% and 0.64%, respectively, compared to the classical KD. Besides, when using ResNet50 as the teacher and MobileNetV1 as the student, ReKD surpasses classical KD with improvements of 2.36% and 1.43% on Top-1 and Top5 accuracy, respectively, establishing a new state-of-the-art result. These findings suggest that a larger number of categories, particularly difficult samples, can provide more valuable knowledge. Furthermore, we hypothesize that the structural differences between the teacher and student may enable the student to learn more generalizable knowledge. Our experimental results confirm this hypothesis, as the extensive range of ImageNet categories allows the student to benefit from rich "dark knowledge" derived from tail categories, contributing to the overall enhancement in performance.

4) Image Classification Results on Tiny-ImageNet: We conduct extensive experiments on the Tiny-ImageNet dataset, evaluating results using both the same and different architectures. The experimental results are presented in Table IV. ReKD has demonstrated superior performance at the logits level compared to other methods. Specifically, ReKD outperforms classical KD, with average gains of 2.07%, 1.96%, 3.72%, 1.93%, and 1.21%, in that order from left to right. These experimental results provide strong evidence that ReKD performs well on fine-grained datasets, such as Tiny-ImageNet.

5) Object Detection Results on MS-COCO: We apply our proposed method to the object detection task. ReKD can be combined with both feature-based methods. While ReviewKD [53] achieved excellent results, the cost of training in features-based distillation remains high. In contrast, our logits-based method has lower computational overhead, similar to logits-based DKD [19], while outperforming it in Table V. ReKD is logits-based and surpasses the classic KD method in multiple evaluation metrics, including mAP, AP50, AP75, APl, APm, and APs. For instance, when using

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TABLE II

TEST ACCURACY (%) OF STUDENTS ON CIFAR-100 VALIDATION. TEACHERS AND STUDENTS ARE IN THE **SAME ARCHITECTURE**. "*" INDICATES THE RESULTS OF OUR IMPLEMENTATION BECAUSE SOME STUDIES ARE NOT DONE EXPERIMENTS ON PARTIAL TEACHER-STUDENT PAIRS. THE BEST RESULTS IN EACH EXPERIMENT ARE EMPHASIZED IN BOLD. EXPERIMENTAL RESULTS OF THE COMPARED METHODS ARE REPORTED IN THEIR ORIGINAL PAPERS OR REPRODUCED FROM PREVIOUS WORK [19], [37], [53]

Distillation Manner	Teacher	ResNet56	ResNet110	ResNet110	ResNet32×4	WRN-40-2	WRN-40-2	VGG13
	Accuracy	72.34	74.31	74.31	79.42	75.61	75.61	74.64
	Student	ResNet20	ResNet20	ResNet32	ResNet8×4	WRN-16-2	WRN-40-1	VGG8
	Accuracy	69.06	69.06	71.14	72.50	73.26	71.98	70.36
Features	AT [39]	70.55	70.22	72.31	73.44	74.08	72.77	71.43
	SP [40]	69.67	70.04	72.69	72.94	73.83	72.43	72.68
	CC [42]	69.63	69.48	71.48	72.97	73.56	72.21	70.71
	VID [52]	70.38	70.16	72.61	73.09	74.11	73.30	71.23
	PKT [44]	70.34	70.25	72.61	73.64	74.54	73.45	72.88
	AB [56]	69.47	69.53	70.98	73.17	72.50	72.38	70.94
	FT [46]	69.84	70.22	72.37	72.86	73.25	71.59	70.58
	FSP [47]	69.95	70.11	71.89	72.62	72.91	71.91*	70.23
	NST [48]	69.60	69.53	71.96	73.30	73.68	72.24	71.53
	FitNet [51]	69.21	68.99	71.06	73.50	73.58	72.24	71.02
	RKD [35]	69.61	69.25	71.82	71.90	73.35	72.22	71.48
	CRD [37]	71.16	71.46	73.48	75.51	75.48	74.14	73.94
	OFD [49]	70.98	70.61*	73.23	74.95	75.24	74.33	73.95
	ICKD [41]	71.76	71.68	73.89	75.25	75.64	74.33	73.88
	WCoRD [50]	71.56	71.57	73.81	**75.95**	75.88	74.73	74.55
	ReviewKD [53]	71.89	71.85*	73.89	75.63	76.12	75.09	74.84
	TaT [57]	71.59	71.70	74.05	75.89	76.06	74.97	74.39
	DPK [58]	**72.37**	**72.44**	**74.89**	75.63	**76.42**	**75.27**	**74.96**
Logits	KD [14]	70.66	70.67	73.08	73.33	74.92	73.54	72.98
	PSKD [59]	71.30	70.91	73.60	75.24	76.01	74.06	74.14
	CTKD [60]	71.19	70.99	73.52	73.70*	75.45	73.93	73.52
	DKD [19]	71.97	70.91*	74.11	76.32	**76.24**	74.81	**74.68**
	ReKD	**72.19**	**71.93**	**74.29**	**76.98**	76.13	**74.84**	74.62
	Δ	+1.53	+1.26	+1.21	+3.65	+1.21	+1.30	+1.64

TABLE III

TOP-1 AND TOP-5 ACCURACY (%) RESULTS ON THE IMAGENET VALIDATION. WE DESIGNATE THE TEACHERS AS **RESNET50** AND **RESNET34**, AND THE STUDENTS AS **MOBILENET (V1)** AND **RESNET18**. D-MANNER IS A DISTILLATION MANNER. Δ INDICATES THE PERFORMANCE IMPROVEMENT OVER KD

D-Manner	Method	Top-1 Accuracy	Top-5 Accuracy	D-Manner	Method	Top-1 Accuracy	Top-5 Accuracy
ResNet34	Teacher	73.31	91.42	ResNet50	Teacher	76.16	92.87
ResNet18	Student	69.75	89.07	MobileNetV1	Student	68.87	88.76
Features	AT [39]	70.69	90.01	Features	AT [39]	70.18	89.68
	OFD [49]	70.81	89.98		OFD [49]	71.25	90.34
	CRD [37]	71.17	90.13		FT [46]	69.88	89.50
	ReviewKD [53]	71.61	90.51		AB [56]	68.89	88.71
	SP [40]	70.62	89.80		RKD [35]	68.50	88.32
	CC [42]	69.96	89.17		ReviewKD [53]	72.56	91.00
	MGD [61]	71.69	90.42		MGD [61]	72.49	90.94
	RKD [35]	70.40	89.78		MGD [62]	71.47	90.35
	NST [48]	70.29	89.53		CRD [37]	71.32	90.41
Logits	KD [14]	70.66	89.88	Logits	KD [14]	70.49	89.92
	PSKD [59]	71.53	90.49		PSKD [14]	-	-
	CTKD [60]	71.32	90.41		WSL [29]	71.52	90.34
	DKD [19]	71.70	90.31		DKD [19]	72.05	91.05
	ReKD	**71.85**	**90.52**		ReKD	**72.85**	**91.35**
	Δ	**+1.19**	**+0.64**		Δ	**+2.36**	**+1.43**

ResNet101 as the teacher and ResNet18 as the student, ReKD achieves performance improvements of 1.38%, 1.97%, 1.16%, 1.94%, 1.35%, 1.65% in the aforementioned metrics, respectively. Besides, when using ResNet101 as the teacher and ResNet50 as the student on the same architecture, ReKD also demonstrated better performance, with average gains of 1.15%, 1.69%, 1.33%, 1.92%, 1.14%, 1.12% in the respective metrics, respectively. Furthermore, experiments with different teacher-student architectures, such as using ResNet50 as the teacher and MobileNetV2 as the student, showed even better performance improvements, with average gains of 2.37%, 3.84%, 2.75%, 2.86%, 2.81%, 2.07% in the mentioned metrics, respectively. Our work focuses on improving the form of logits for mining dark knowledge and the experimental results demonstrate that we achieve superior performance at the logit distillation. However, as depicted in Table V, when

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TABLE IV
ACCURACY* AND DKD* REPRESENT THE RESULT OF OUR IMPLEMENTATION. EXPERIMENTAL RESULTS OF THE COMPARED METHODS ARE REPORTED IN THEIR ORIGINAL PAPERS OR REPRODUCED FROM PREVIOUS WORK [63]

Distillation Manner	Teacher	ResNet56	ResNet110	VGG13	WRN-40-2	VGG13
	Accuracy	58.34	58.46	60.09	61.26	60.09
	Accuracy*	56.56	59.04	60.22	60.43	60.22
	Student	ResNet20	ResNet20	VGG8	WRN-16-2	MobileNetV2
	Accuracy	52.66	51.89	56.03	57.17	57.73
Features	AT [39]	54.39	54.57	58.85	59.39	60.84
	SP [40]	54.23	54.38	58.78	57.63	61.90
	CC [42]	54.22	54.26	58.18	58.83	61.32
	VID [52]	53.89	53.94	58.55	58.78	60.84
	PKT [44]	54.29	54.70	58.87	59.19	61.90
	FT [46]	53.90	54.46	58.87	58.85	61.78
	NST [48]	53.66	53.82	58.85	59.07	60.59
	FitNet [51]	54.43	54.04	58.33	58.88	61.37
	RKD [35]	53.95	53.88	58.58	59.31	61.19
	CRD [37]	55.04	54.69	58.88	59.42	61.63
	SAKD [63]	55.06	55.28	59.53	59.87	**62.29**
Logits	KD [14]	53.04	53.40	57.33	59.16	60.02
	DKD* [19]	54.92	55.13	60.05	59.63	59.63
	ReKD	**55.11**	**55.36**	**61.05**	**61.09**	61.23
	Δ	+2.07	+1.96	+3.72	+1.93	+1.21

compared to features-based methods (e.g., ReviewKD [53]), ReKD based on logits distillation is suboptimal. Therefore, as an optimization strategy to further enhance performance, combining ReKD with features-based methods can achieve better performance. As illustrated in Table V, ReKD significantly improves the student's performance, surpassing even the state-of-the-art ReviewKD. This finding confirms that ReKD can effectively combine with features-based methods.

6) Instance Segmentation Results on MS-COCO: ReKD is not only applicable to classification tasks but also to more challenging instance segmentation tasks. By combining features using the ReviewKD method, ReKD can learn the discriminative power of deep features in both the same and different model architectures. As shown in Table VI, ReKD achieves superior results compared to ReviewKD when combining features. These experimental results validate the effectiveness of ReKD in instance segmentation tasks as well.

B. Ablation Studies

In this section, we first conduct an overall ablation experiment to explore the impact of TISD and NTID on different datasets and models. The results are summarized in Table VII. We then investigate the effectiveness of TISD and NTID components, which are presented in Tables VIII and IX. Finally, we analyze the impact of each component individually.

1) Effect of TISD and NTID: In this paper, we introduce two novel components designed for knowledge transfer. To evaluate the effectiveness of these components, we adopt both homogeneous and heterogeneous teacher-student model architectures and train student models with each component separately on various task-specific datasets. As demonstrated in Table VII, the ablation results indicate that employing either TISD or NTID alone surpasses the performance of the classical Knowledge Distillation method. This evidence further validates that combining both components can enhance the student's performance. Therefore, we conclude that the two components we designed are both effective and reliable.

In addition, for simpler datasets like CIFAR100, as depicted in Fig. 3 (c) and (d), knowledge distillation for tail categories offers a significant advantage over head categories. However, for more challenging datasets like TinyImageNet, as shown in Fig. 3 (a) and (b), the distillation for head categories proves to be more effective. This phenomenon may be attributed to the challenging samples providing richer knowledge of the head categories. When the knowledge across categories is more similar, it allows for mutual referencing, potentially compensating for erroneous target predictions. Conversely, simpler samples may constrain the distillation effects for tail categories due to dominant categories (e.g., top-2, top-3 categories) with higher probabilities of existing inter-class competition compared to challenging samples. As a result, different datasets also affect TISD and NTID dominance for distillation performance.

2) Effect of TISD: KD typically employs softmax with a temperature coefficient $\mathcal{T}$ to soften label information. A higher temperature value yields a smoother probability distribution in the model's output, while a lower temperature value makes the probability distribution closer to the "one-hot" label. However, it is a known fact that increasing the temperature reduces the feature distinction of the knowledge of head categories, as it introduces knowledge from tail categories. This issue may hinder the effective utilization of similar information. Therefore, it is essential to distinguish the head categories information to improve the classification performance of the student.

To address this issue, we introduce the TISD component that primarily captures the head categories information effectively. The teacher emphasizes imparting this knowledge while avoiding the incorporation of additional noisy information. Additionally, the TISD component focuses on transferring these more relevant feature representations to the student. In order to investigate the role of TISD and validate its effectiveness, we present the following detailed experimental analysis (All ablation experiments were conducted with $K = 7$).

Similar to TISD, classical KD essentially transfers the knowledge of head categories and ignores this knowledge when achieving optimal performance. In order to maximize the utilization of knowledge of head categories while minimizing the introduction of additional noisy categories, we propose TISD component. As reported in Table VIII-(a) and Table VIII-(b), TISD exhibits comparable or superior performance to classical KD, with average gains of 1.31%, 0.50%, and 0.04% in ResNet32 × 4 vs ResNet8 × 4, ResNet32 × 4 vs ShuffleNetV2, and WRN-40-2 vs WRN-16-2 on the CIFAR-100 [66] dataset, respectively. These experimental results validate that TISD effectively leverages the knowledge of head categories with inter-class similarity information. Furthermore, we suggest that classical KD mainly relies on knowledge of head categories as dark knowledge to achieve optimal performance. It rarely takes advantage of additional knowledge of head categories to improve classification performance, even when using the same temperature coefficient for softening as TISD.

TISD is particularly adept at capturing the knowledge of head categories, especially for inter-class knowledge with higher similarity. The fine-grained dataset categories (e.g., CUB-200-2011 [67], Tiny-ImageNet) exhibit a higher

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TABLE V

THE TEST ACCURACY (%) OF STUDENTS ON MS-COCO2017 DATESET. THE BEST RESULTS IN EACH EXPERIMENT ARE EMPHASIZED IN BOLD. RESNET110, RESNET50, RESNET18, AND MOBILENETV2 ARE DENOTED AS R101, R50, R18, AND MV2, RESPECTIVELY. T&S IS TEACHER AND STUDENT

T&S Model	Distillation Types	Methods	mAP	AP50	AP75	APl	APm	APs
R101& R18	Teacher	R101-FPN(T)	42.04	62.48	45.88	54.60	45.55	25.22
	Student	R18-FPN(S)	33.26	53.61	35.26	43.16	35.68	18.96
	Features	FitNet [51]	34.13	54.16	36.71	44.69	36.50	18.88
		FGFI [64]	35.44	55.51	38.17	47.34	38.29	19.04
		ReviewKD [53]	36.75	56.72	34.00	49.58	39.51	19.42
		ReviewKD+ReKD	**36.85**	**57.35**	**36.69**	**49.64**	**39.73**	**19.76**
		Δ	**+3.59**	**+3.74**	**+1.43**	**+6.48**	**+4.05**	**+0.80**
	Logits	KD [14]	33.97	54.66	36.62	44.14	36.67	18.71
		CTKD [60]	34.56	55.43	36.91	45.07	37.21	19.08
		DKD [19]	35.05	56.60	37.54	-	-	-
		ReKD	**35.35**	**56.63**	**37.78**	**46.08**	**38.02**	**20.36**
		Δ	**+1.38**	**+1.97**	**+1.16**	**+1.94**	**+1.35**	**+1.65**
R101& R50	Teacher	R101-FPN(T)	42.04	62.48	45.88	54.60	45.55	25.22
	Student	R50-FPN(S)	37.93	58.84	41.05	49.10	41.14	22.44
	Features	FitNet [51]	38.76	59.62	41.80	50.70	42.20	22.32
		FGFI [64]	39.44	60.27	43.04	51.97	42.51	22.89
		ReviewKD [53]	40.36	60.97	44.08	52.87	43.81	23.60
		ReviewKD+ReKD	**40.57**	**61.48**	**44.19**	**53.17**	**44.09**	**23.91**
		Δ	**+2.64**	**+2.64**	**+3.14**	**+4.07**	**+2.95**	**+1.47**
	Logits	KD [14]	38.35	59.41	41.71	49.48	41.80	22.73
		DKD [19]	39.25	60.90	42.73	-	-	-
		ReKD	**39.50**	**61.10**	**43.04**	**51.40**	**42.94**	**23.85**
		Δ	**+1.15**	**+1.69**	**+1.33**	**+1.92**	**+1.14**	**+1.12**
R50& MV2	Teacher	R50-FPN(T)	40.22	61.02	43.81	51.98	43.53	24.16
	Student	MV2-FPN(S)	29.47	48.87	30.90	38.86	30.77	16.33
	Features	FitNet [51]	30.20	49.80	31.69	39.69	31.64	16.39
		FGFI [64]	31.16	50.68	32.92	42.12	32.63	16.73
		ReviewKD [53]	33.71	53.15	36.13	**46.47**	35.81	16.77
		ReviewKD+ReKD	**34.24**	**54.91**	**36.56**	46.06	**36.56**	**18.71**
		Δ	**+4.77**	**+6.04**	**+5.66**	+7.20	**+5.79**	**+2.38**
	Logits	KD [14]	30.13	50.28	31.35	39.56	31.91	16.69
		CTKD [60]	30.39	52.34	33.30	41.06	33.56	18.15
		DKD [19]	32.34	53.77	34.01	-	-	-
		ReKD	**32.50**	**54.12**	**34.10**	**42.42**	**34.72**	**18.76**
		Δ	**+2.37**	**+3.84**	**+2.75**	**+2.86**	**+2.81**	**+2.07**

TABLE VI

THE TEST ACCURACY (%) OF STUDENTS FOR INSTANCE SEGMENTATION RESULTS ON MS-COCO2017 DATASET. WE CHOOSE MASK-RCNN [6] AS THE METHOD FOR NETWORK TRAINING. R101-FPN DENOTES RESNET101 COMBINED WITH THE FPN NETWORK FOR FEATURE EXTRACTION

Model	Distillation Methods	mAP	AP50	AP75	APl	APm	APs
Teacher: R101-FPN	-	38.63	60.45	41.28	55.29	41.33	19.48
Student: R18-FPN	-	31.25	51.07	33.10	45.53	32.80	14.18
	+ ReviewKD [53]	33.62	53.91	35.96	50.30	35.31	**15.03**
	+ ReviewKD [53]+ ReKD	**33.87**	**54.26**	**36.06**	**50.69**	**35.37**	**15.03**
	Δ	**+2.62**	**+3.19**	**+2.96**	**+5.16**	**+2.57**	**+0.85**
Teacher: R101-FPN	-	38.63	60.45	41.28	55.29	41.33	19.48
Student: R50-FPN	-	35.24	56.32	37.49	50.34	37.71	17.16
	+ReviewKD [53]	36.98	58.13	39.60	53.19	39.57	17.50
	+ReviewKD [53]+ ReKD	**37.17**	**58.74**	**39.72**	**53.67**	**39.65**	**18.04**
	Δ	**+1.93**	**+2.42**	**+2.23**	**+3.33**	**+1.94**	**+0.88**
Teacher: R50-FPN	-	37.17	58.60	39.88	53.30	39.49	18.63
Student: MV2-FPN	-	28.37	47.19	29.95	41.70	29.01	12.09
	+ ReviewKD [53]	31.56	50.70	33.44	47.39	32.44	12.76
	+ ReviewKD [53]+ ReKD	**31.97**	**52.14**	**33.83**	**47.76**	**33.08**	**14.45**
	Δ	**+3.60**	**+4.95**	**+3.88**	**+6.06**	**+4.07**	**+2.36**

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TABLE VII

THE ABLATION EXPERIMENTS OF TISD AND NTID COMPONENTS ON DIFFERENT DATASETS AND MODELS. THE ACCURACY OF COCO IS MAP

Dataset	Model	TISD	NTID	Accuracy	Δ
CIFAR-100	T:ResNet32x4			79.42	-
	S:ResNet8x4			72.50	-
		√		74.47	+1.97
			√	74.64	+2.14
		√	√	**76.98**	+4.48
Tiny-ImageNet	T:VGG13			60.22	-
	S:MobileNetV2			57.73	-
		√		60.31	+2.58
			√	59.63	+1.90
		√	√	**61.00**	+3.27
ImageNet-1K	T:ResNet34			73.31	-
	S:ResNet18			69.75	-
		√		71.70	+1.95
			√	71.54	+1.79
		√	√	**71.85**	+2.10
	T:ResNet50			76.16	-
	S:MobileNetV1			68.87	-
		√		72.71	+3.84
			√	72.27	+3.40
		√	√	**72.85**	+3.98
MS-COCO2017	T:ResNet101			42.04	-
	S:ResNet18			33.26	-
		√		34.05	+0.79
			√	34.52	+1.26
		√	√	**35.35**	+2.09
	T:ResNet50			40.22	-
	S:MobileNetV2			29.47	-
		√		30.53	+1.06
			√	31.46	+1.99
		√	√	**32.50**	+3.03

TABLE VIII

THE ABLATION EXPERIMENTS OF TISD COMPONENT STUDY THE IMPACT ON DIFFERENT MODELS AND CIFAR-100 DATASET."W/" DENOTES WITH AUTOAUGMENT [65]."*" REPRESENTS THE RESULT WE REPRODUCED AND DKD [19] REPORTS 75.33 AND 77.98 FOR RESNET8 × 4 AND SHUFFLENETV1 RESPECTIVELY WITH AUTOAUGMENT

	Model	Classical KD	TISD	Accuracy	Δ
(a)	T:ResNet32x4			79.42	-
	S:ResNet8x4			72.50	-
		√		73.33	-
			√	**74.64**	+1.31
	S:ShuffleNetV2			71.82	-
		√		74.45	-
			√	**74.95**	+0.50
(b)	T:WRN-40-2			75.61	-
	S:WRN-16-2			73.26	-
		√		74.92	-
			√	**74.96**	+0.04
	S:ShuffleNetV1			70.50	-
		√		**74.83**	-
			√	74.67	-0.16
	Model	Classical KD	TCKD	TISD	Accuracy	Δ
(c)	T:ResNet32x4				81.24	-
	S:ResNet8x4	√ (W/)			75.66	-
			√ (W/)		73.21*	-
				√ (W/)	**76.77**	+1.11
	S:ShuffleNetV1	√ (W/)			77.13	-
			√(W/)		75.05*	-
				√ (W/)	**77.74**	+0.61

degree of inter-class similar categories than the CIFAR-100 dataset. Therefore, it is more challenging to classify accurately. We experimentally analyze that TISD significantly outperforms classical KD on more fine-grained datasets. As depicted in Table IX-(a), TISD significantly outperforms classical KD on the Tiny-ImageNet dataset, with average gains of 1.51%

TABLE IX

EVALUATING THE EFFECTIVENESS OF TISD AND NTID COMPONENTS ON STUDENT'S PERFORMANCE. ACCURACY* REPRESENTS THE RESULT OF OUR IMPLEMENTATION

	Distillation Dataset	Teacher / Student	ResNet56 / ResNet20	WRN-40-2 / WRN-16-2
(a)	Tiny-ImageNet	T: Accuracy*	56.56	60.43
		S: Accuracy	52.66	57.17
		Classical KD	53.04	59.16
		TISD	**54.55**	**59.90**
		Δ	+1.51	+0.74
		NTID	**55.52**	**59.84**
		Δ	+1.48	+0.68
(b)	CUB-200-2011	T: Accuracy	62.08	61.20
		S: Accuracy	48.34	42.95
		Classical KD	59.79	56.43
		TISD	**61.49**	**58.24**
		Δ	+1.70	+1.81
		NTID	**61.94**	**57.37**
		Δ	+2.15	+0.94
(c)	ImageNet-1K	Teacher	ResNet34	ResNet50
		Student	ResNet18	MobileNetV1
		T: Accuracy	73.31	76.16
		S: Accuracy	69.75	68.87
		Classical KD	70.66	70.50
		NTID	**71.54**	**72.27**
		Δ	+0.88	+1.77

TABLE X

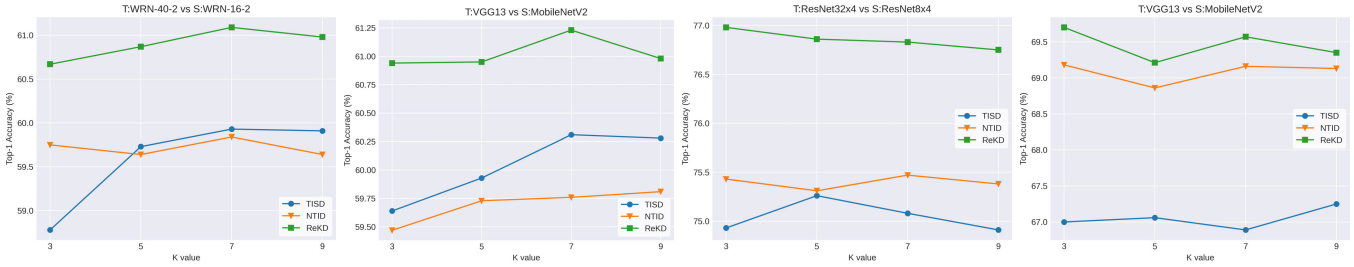
PERFORMANCE IMPACT OF VARIOUS K VALUES ON TISD AND NTID IN COMPARISON WITH TCKD AND NCKD ON THE TINY-IMAGENET DATASET

Dataset	Model	TCKD	TISD			
Tiny-Imagenet	Teacher: WRN-40-2 vs Student: WRN-16-2	-	K=3	K=5	K=7	K=9
		56.39	58.78	59.73	59.93	59.91
		NCKD	NTID			
		-	K=3	K=5	K=7	K=9
		59.42	59.75	59.64	59.84	59.64
Dataset	Model	TCKD	TISD			
Tiny-Imagenet	Teacher: VGG13 vs Student: MobileNetV2	-	K=3	K=5	K=7	K=9
		56.09	59.64	59.93	60.31	60.28
		NCKD	NTID			
		-	K=3	K=5	K=7	K=9
		59.34	59.47	59.73	59.76	59.81

and 0.74% in ResNet56 vs ResNet20, and WRN-40-2 vs WRN-16-2, respectively. Besides, as offered in Table IX-(b), the performance of the CUB-200-2011 dataset improves with average gains of 1.70% and 1.81% in ResNet56 vs ResNet20, and WRN-40-2 vs WRN-16-2, respectively. These results suggest that TISD indeed captures the knowledge of head categories, especially for inter-class knowledge with higher similarity on fine-grained datasets. Thus, TISD is more conducive to improving student performance compared to classical KD on the CIFAR-100 dataset, as illustrated in Tables VIII.

TISD has the ability to narrow the gap between the student and teacher. The experimental details can be found in the Supplementary Material. When there is a significant gap between the student and teacher, the teacher's predicted probability distribution (red curve) is often sharper and stronger than the student's (green curve), particularly for challenging training samples. If the student's probability distribution directly matches the teacher's, it may not learn additional discriminative category information due to limited capacity, which

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(a) Tiny-ImageNet (b) Tiny-ImageNet (c) CIFAR100 (d) CIFAR100

Fig. 3. Ablation about different values of parameter K affect the performance of ReKD, TISD and NTID in different datasets and model architectures.

hinders network parameter optimization. Therefore, we utilize TISD to capture the head categories' information, represented by the probability distribution (blue curve), while removing some low-probability category information that may be noisy. This transformation makes the probability distribution (changing the green curve to the blue curve) slightly steeper, enabling the student to concentrate on learning primary knowledge of head categories. Our experimental results show that TISD can achieve a +3.89% (59.20% vs 63.09%) performance gain compared to the teacher assistant method on the CIFAR-100 dataset. These results verify that TISD effectively narrows the gap between the teacher and student. The approach emphasizes learning the knowledge of head categories, which is most representative and discriminative while reducing the burden of acquiring additional categories of knowledge with low probability.

3) Effect of NTID: To investigate the value of knowledge of tail categories, we hypothesize that difficult samples provide richer dark knowledge than simple ones, thereby enhancing the learning of knowledge of tail categories. Thus, we conduct experiments on different datasets with numerous classes. Firstly, we evaluate the effectiveness of NTID on the fine-grained CUB-200-2011 [67] bird dataset. It poses challenges due to its high similarities in bird shape, features, colors, and other characteristics. As offered in Table IX-(b), our experimental results reveal that NTID significantly outperforms classical KD, with average gains of 2.15% and 0.94% in ResNet56 vs ResNet20 and WRN-40-2 vs WRN-16-2, respectively. This confirms that fine-grained datasets also contain rich knowledge of tail categories that can be transferred to student learning. In addition, we assess the effectiveness of NTID on the challenging ImageNet-1K dataset. As reported in Table IX-(c), we observe that the knowledge of tail categories plays a crucial role in improving the student's performance, with average gains of 0.88% and 1.77% in ResNet34 vs ResNet18 and ResNet50 vs MobileNetV1, respectively.

In summary, the knowledge of tail categories enhances network generalization performance by learning valuable knowledge. It helps the student network optimize and converge more quickly.

4) Effect of Parameter K on Performance: For a fair comparison across different datasets and teacher-student model architectures, we determined the optimal K value from ReKD's performance. As illustrated in Fig. 3 (a) and (b), for the Tiny-ImageNet dataset, the performance of ReKD (represented by the green curve) peaks when $K = 7$. Conversely, for the CIFAR100 dataset, as shown in Fig. 3 (c) and (d), $K = 3$ is the optimal choice for the ReKD. It is notable that more challenging datasets tend to have a relatively larger optimal K value. In summary, to thoroughly harness knowledge from both the head and tail categories and ensure optimal performance, the optimal K value varies depending on the dataset.

C. Comparison of DKD and ReKD

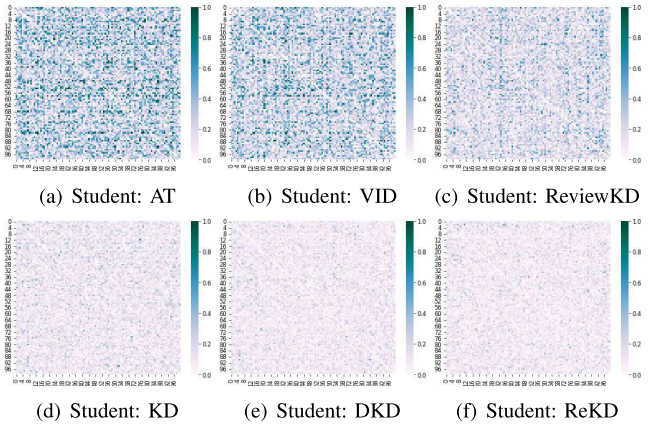
1) Comparison of TCKD and TISD: In the realm of knowledge distillation, head categories distillation plays a pivotal role in transferring knowledge, enabling it to better comprehend the distinguishing characteristics and relationships among different categories

Firstly, TISD proves valuable when the prediction for the target category is mistaken, TISD can leverage related or similar head categories to guide its learning, compensating for the discrepancies in the ground-truth prediction. Secondly, TCKD demonstrates significant improvements under certain conditions when dealing with more challenging samples, such as those with noise or subjected to data augmentation. However, these conditions might limit its overall efficacy, especially when such conditions are not explicit. In contrast, TISD can effectively deal with these challenging scenarios by fine-tuning the hyper-parameter K. This adjustment results in a performance increase of 3.54% (56.39% vs 59.93%) and 4.22% (56.09% vs 60.31%), as presented in Table X. In addition, the performance of TISD using data augmentation technology has been significantly improved compared to TCKD, as shown in Table VIII-(c). Hence, TISD exhibits a more pronounced advantage over TCKD, verifying its effectiveness.

2) Comparison of NCKD and NTID: The successful implementation of knowledge distillation depends on the effective distillation of non-target categories [19], [25]. However, intricate or rarely encountered categories often pose challenges for transferring knowledge in NCKD. This complexity arises because these non-target categories often exhibit shared similarities. There exists a potential for inadvertently introducing misleading information in the absence of proper comparisons with the target category. Compared to NCKD, NTID focuses on a more confined range of tail categories, which aids in reducing inter-category competition and attenuating noise-induced perturbations.

It is worth emphasizing the distinct strategies that DKD and ReKD employ to handle non-target categories. Specifically, firstly, the decoupling of DKD focuses on non-target categories distillation. However, incomplete decoupling may

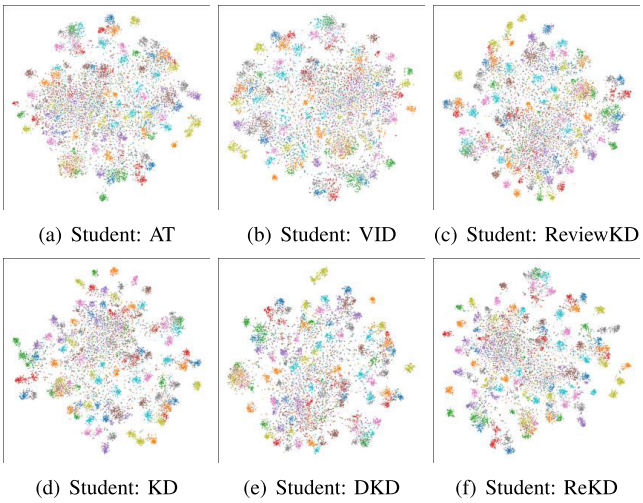
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(a) Student: AT (b) Student: VID (c) Student: ReviewKD

(d) Student: KD (e) Student: DKD (f) Student: ReKD

Fig. 4. The correlations between class logits of a teacher and student. We adopt ResNet56 as a teacher and ResNet20 as a student on the CIFAR-100 dataset. The final presentation is the result of the student.



(a) Student: AT (b) Student: VID (c) Student: ReviewKD

(d) Student: KD (e) Student: DKD (f) Student: ReKD

Fig. 5. t-SNE of features learned by the student. We adopt ResNet32 $\times$ 4 as a teacher and ResNet8 $\times$ 4 as a student on the CIFAR-100 dataset.

impede the seamless knowledge transfer from these categories. In contrast, ReKD provides an avenue to adjust the suppression level using the parameter K. For instance, opting for a smaller K value, like $K = 2$, may result in classes with higher probabilities suppressing the knowledge transfer from tail categories. But increasing K might effectively mitigate this suppression in ReKD. Secondly, tail categories, often associated with low probabilities, contain complex and challenging information that may be novel to the student model. Assimilating these categories can bolster the model's generalization capability. We can utilize this critical information more effectively by implementing tail categories knowledge distillation. For instance, when dealing with more complex datasets such as Tiny-ImageNet, our method can achieve significantly improved performance compared to DKD, with a performance increase of 0.42% (59.42% vs 59.84%) and 0.42% (59.34% vs 59.76%) across different model architectures when K is 7, as shown in Table X.

3) The Advantages of ReKD: First, ReKD optimizes knowledge transfer by introducing the hyper-parameter K; ReKD addresses the performance difference compared to DKD's TCKD and NCKD (highlighted in Table I from [19]), balancing the importance of head and tail categories for performance optimization. Second, while DKD primarily differentiates between categories, ReKD assimilates teacher model predictions holistically, ensuring comprehensive learning from categories the teacher model is likely to predict and leverage knowledge from tail categories. Third, by adapting the hyper-parameter K to different datasets, ReKD offers nuanced knowledge transfer. In complex datasets like Tiny-ImageNet, as shown in Fig 3 (a) and (b), ReKD demonstrates superior performance by tuning K, especially in challenging scenarios with unseen data in flexibility.

D. Quantitative Analysis

This subsection employs the t-SNE technique [68], [69] and correlation matrices to visualize our experimental results, which demonstrate the effectiveness of our proposed method in comparison with other existing methods.

1) Visualization of Correlation Matrices: We visualize the discrepancy between the correlation matrices of the teacher and student at the logits layer in Fig. 4. It can be observed that ReKD exhibits a smaller difference, indicating a closer alignment between the teacher's performance and the student's. This suggests that student learns more detailed information from the teacher, which is beneficial for improving the student's classification performance.

2) Visualization of t-SNE: To further analyze the relationship between the teacher and student, we utilize the t-SNE technique [68] to visualize the experimental results. As depicted in Fig. 5(a-c), ReKD generates larger distances between categories and more compact cluster centers for each category compared to features-based methods. As illustrated in Fig. 5(f), we demonstrate that our proposed ReKD enables the student to learn more discriminative features when compared to the logits-based methods, such as KD in Fig. 5(d) and DKD in Fig. 5(e).

VI. Conclusion

In this paper, we propose ReKD, an effective knowledge distillation method that comprises two components: TISD and NTID. Our method surpasses existing approaches by better leveraging the knowledge of both head and tail categories. It is easy to implement, incurs no extra computational cost, and can further improve performance when combined with features-based methods. Moreover, we provide a theoretical derivation of the upper bound of the loss function. By substituting the original decoupled weighting coefficients with two adjustable parameters, we enable the student network to learn knowledge more flexibly. Experimental results show that our method achieves state-of-the-art performance on the CIFAR-100, Tiny-ImageNet, and ImageNet-1K datasets for image classification, as well as the MS-COCO dataset for object detection and instance segmentation. In future work, we aim to explore the decoupling of feature maps in intermediate layers for image classification, instance segmentation, and object detection tasks. This will enable students to learn more knowledge and overcome the limitations of the performance of existing knowledge distillation methods.

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